THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum of coefficients in the balanced equation $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$ is: 1

(a) 10 **(b)** 9 **(c)** 8 **(d)** 12

2. The pH of a solution is 5. If its pH becomes 3, its H^+ concentration: 1

(a) decreases 2 times **(b)** decreases 100 times

(c) increases 100 times **(d)** increases 2 times

3. Which of the following will not react with dilute HCl? 1

(a) magnesium **(b)** iron **(c)** zinc **(d)** silver

4. The number of covalent bonds in butane C_4H_{10} is: 1

(a) 13 **(b)** 10 **(c)** 14 **(d)** 12

5. In $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$, the substance reduced is: 1

(a) Al **(b)** Fe **(c)** Fe_2O_3 **(d)** Al_2O_3

6. The solid residue obtained on heating washing soda crystals is: 1

(a) anhydrous sodium carbonate **(b)** sodium hydroxide

(c) sodium chloride **(d)** sodium bicarbonate

7. Which of the following is correct about ionic compounds? 1

(a) They are gaseous **(b)** They are insoluble in water

(c) They are soft **(d)** They have high melting points

8. CH_3CH_2OH on heating with excess conc. H_2SO_4 at 443 K gives: 1

(a) $CH_2{=}CH_2$ **(b)** CH_3COOH **(c)** CH_3CH_3 **(d)** CH_3CHO

9. The colour of copper sulphate solution fades when an iron nail is dipped in it because: 1

(a) $FeSO_4$ (pale green) is formed **(b)** iron turns blue

(c) copper sulphate evaporates **(d)** water is formed

10. A student tests an unknown solution with litmus and gets no change. The solution is most likely: 1

(a) acidic **(b)** strongly acidic **(c)** basic **(d)** neutral

11. Iron articles are painted to prevent rusting because paint: 1

(a) makes iron reactive **(b)** reacts with iron

(c) keeps air and water away **(d)** makes iron heavy

12. Which of these is NOT a property of carbon? 1

(a) tetravalency **(b)** catenation
(c) forms ionic bonds with chlorine **(d)** forms multiple bonds

13. The coefficient of O_2 in the balanced equation $C_3H_8 + O_2 \longrightarrow CO_2 + H_2O$ is: 1

(a) 5 **(b)** 4 **(c)** 7 **(d)** 3

14. Which of the following will have the highest H^+ ion concentration? 1

(a) pH 1 **(b)** pH 4 **(c)** pH 10 **(d)** pH 7

15. A metal M reacts with cold water giving hydrogen and a base. M could be: 1

(a) copper **(b)** sodium **(c)** silver **(d)** gold

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Differentiate between a physical change and a chemical change with one example each. 2

17. Differentiate between a strong acid and a weak acid with one example each. 2

18. Why does calcium float while reacting with water? 2

19. Why does carbon form covalent compounds only and not ionic ones? 2

20. Balance $Al + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2$ and name the gas evolved. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Balance $Al + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2$ and state whether it is a displacement or a combination reaction. 3

22. A student adds Mg ribbon to three test tubes with HCl, CH_3COOH and distilled water. Predict the observation in each and explain. 3

23. Explain with equations how aluminium reacts with oxygen and what protects it from further corrosion. 3

24. Calculate the number of covalent bonds in 3
(a) hexane C_6H_{14} and
(b) $C_{10}H_{22}$.

OR

Explain why the thermal decomposition of limestone needs continuous heating. Write the equation and one use of the solid product. 3

25. Explain the preparation of sodium hydroxide, naming the electrodes and products. 3

OR

Explain the concentration of ore steps for a zinc sulphide ore. 3

MASTERMIND

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Classify each as combination, decomposition, displacement or double displacement: **5**

(a) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$

(b) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$

(c) $BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2\,HCl$

(d) $2\,Mg + O_2 \longrightarrow 2\,MgO$

(e) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$.

OR

A student has four unlabelled test tubes with HCl, NaOH, NaCl and Na_2CO_3 solutions. Suggest tests to identify each. **5**

27. Explain, giving equations, the reaction of metals with **5**

(a) oxygen

(b) water

(c) acids using one metal for each.

OR

Explain tetravalency, catenation and covalent bonding of carbon with three reasons for the large number of carbon compounds. **5**

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND